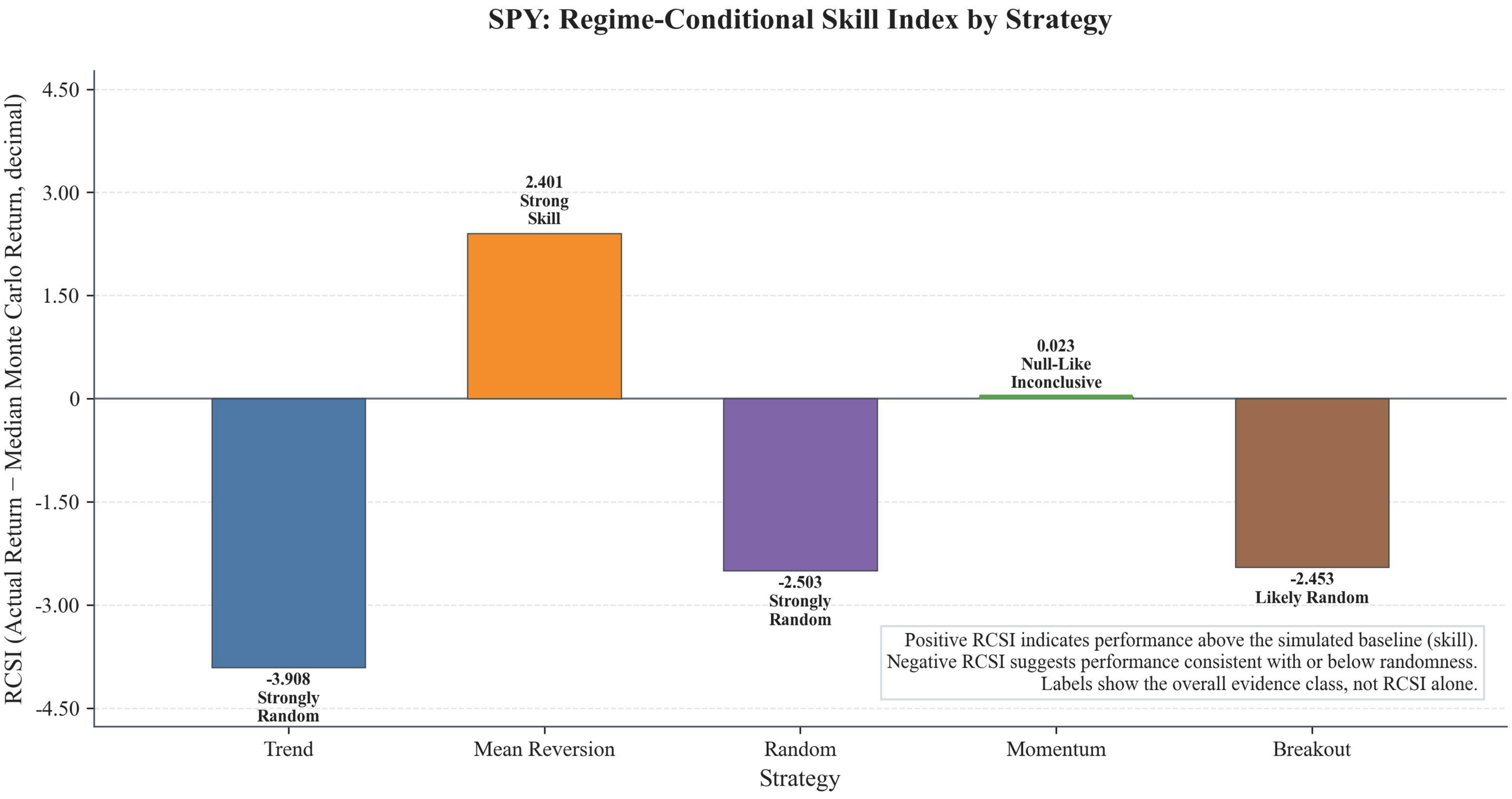
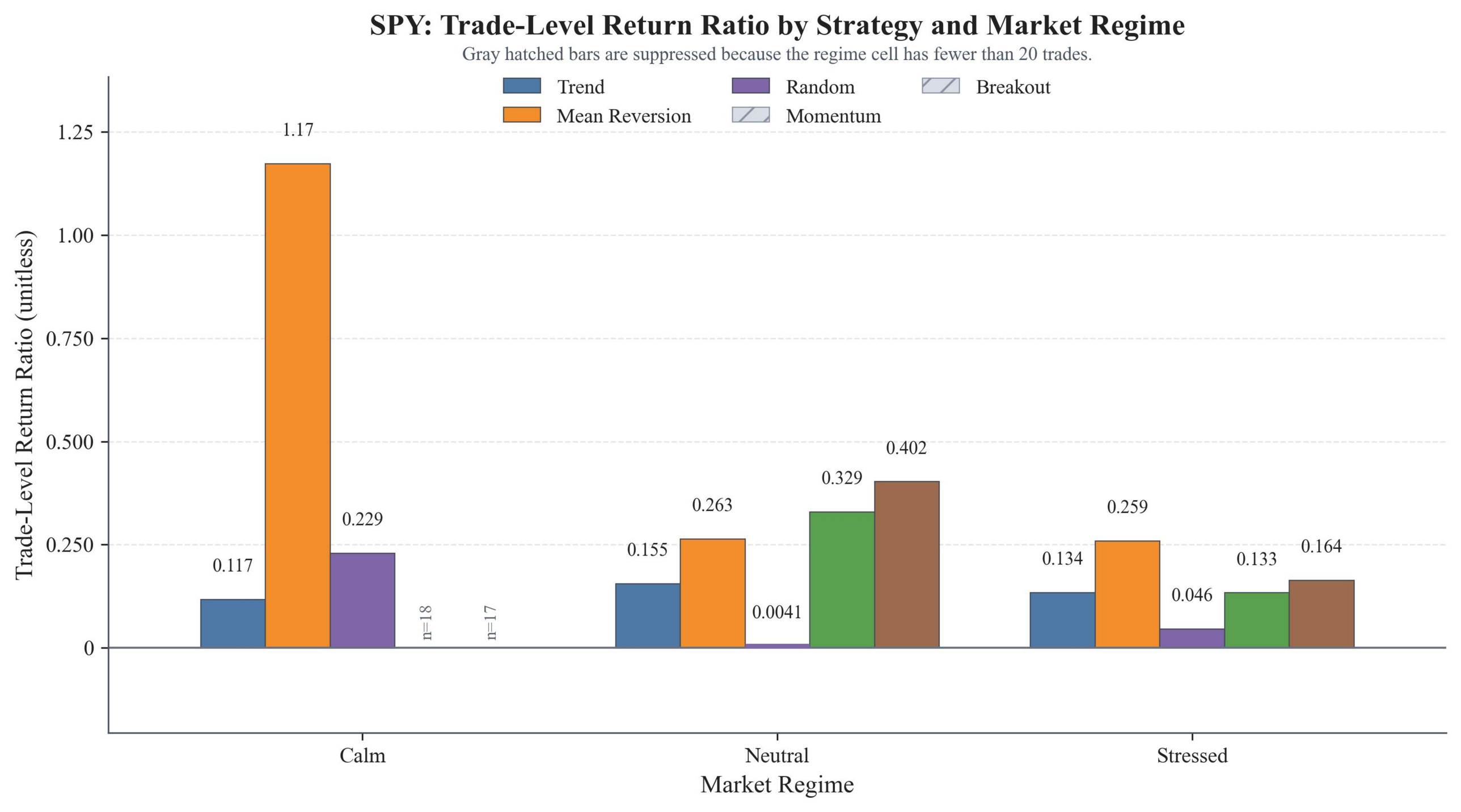
SPY: Strategy Evidence and Confidence Overview

Evidence labels use scale-free statistics: p-value, percentile, normalized RCSI, robustness, and stability.

Strategy	Evidence	Verdict	Confidence	p-value	Prom.	Pctile	RCSI_z
Trend	Strongly Random	Luck	High	0.934	0.03	6.6	-1.14
Mean Reversion	Strong Skill	Skill	High	0.009	2.02	99.0	3.18
Random	Strongly Random	Luck	High	0.990	0.00	1.0	-1.42
Momentum	Null-Like / Inconclusive	Inconclusive	Low	0.504	0.30	49.6	-0.21
Breakout	Likely Random	Luck	Moderate	0.823	0.08	17.7	-0.85

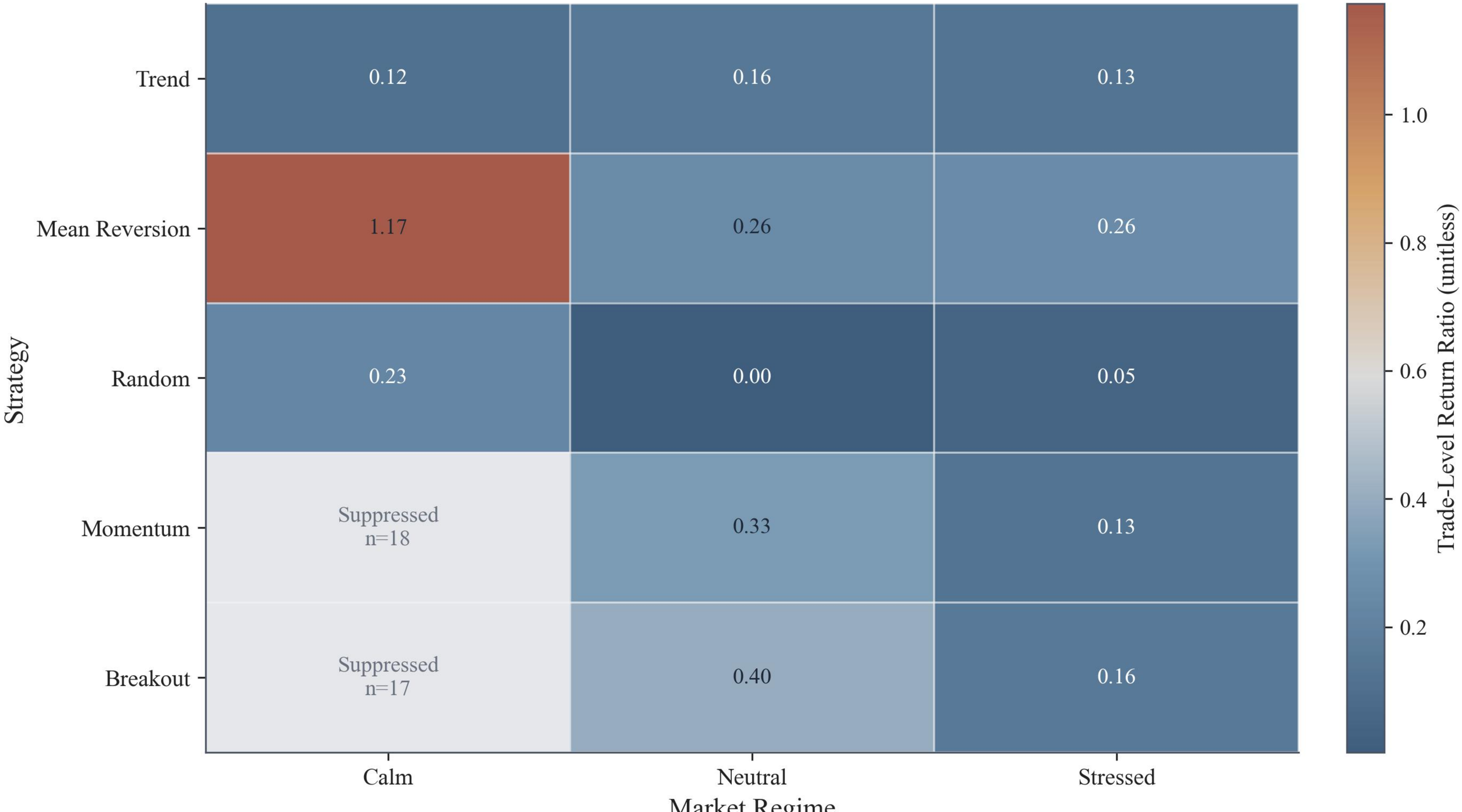
Interpretation:
Prom. = -log10(p), so larger values indicate rarer outcomes under the null.
Pctile and RCSI_z provide scale-free evidence and are more comparable across tickers.
Benchmark context appears in the comparison table and equity curve rather than this verdict summary.


SPY: Regime-Conditional Skill Index by Strategy
RCSI (Actual Return − Median Monte Carlo Return, decimal)
4.50
3.00
1.50
0
-1.50
-3.00
-4.50
2.401
Strong
Skill
0.023
Null-Like
Inconclusive
-3.908
Strongly
Random
-2.503
Strongly
Random
-2.453
Likely Random
Positive RCSI indicates performance above the simulated baseline (skill).
Negative RCSI suggests performance consistent with or below randomness.
Labels show the overall evidence class, not RCSI alone.
Trend
Mean Reversion
Random
Momentum
Breakout
Strategy

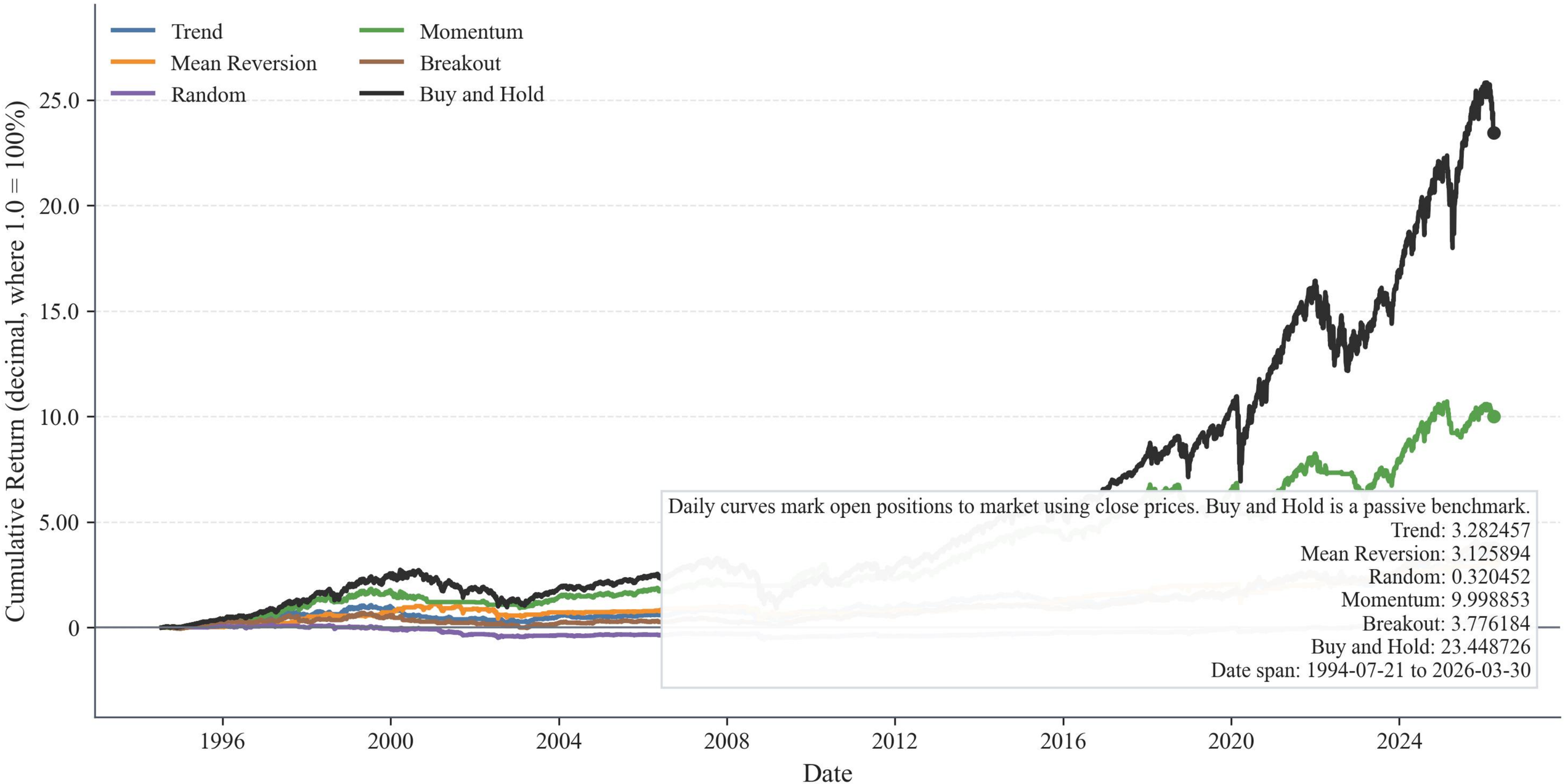

SPY: Trade-Level Return Ratio by Strategy and Market Regime
Gray hatched bars are suppressed because the regime cell has fewer than 20 trades.
Trend
Mean Reversion
Random
Momentum
Breakout
Trade-Level Return Ratio (unitless)
1.25
1.00
0.750
0.500
0.250
0
1.17
0.117
0.229
n=18
n=17
0.155
0.263
0.0041
0.329
0.402
0.134
0.259
0.046
0.133
0.164
Calm
Neutral
Stressed
Market Regime

SPY: Strategy Performance Across Market Regimes

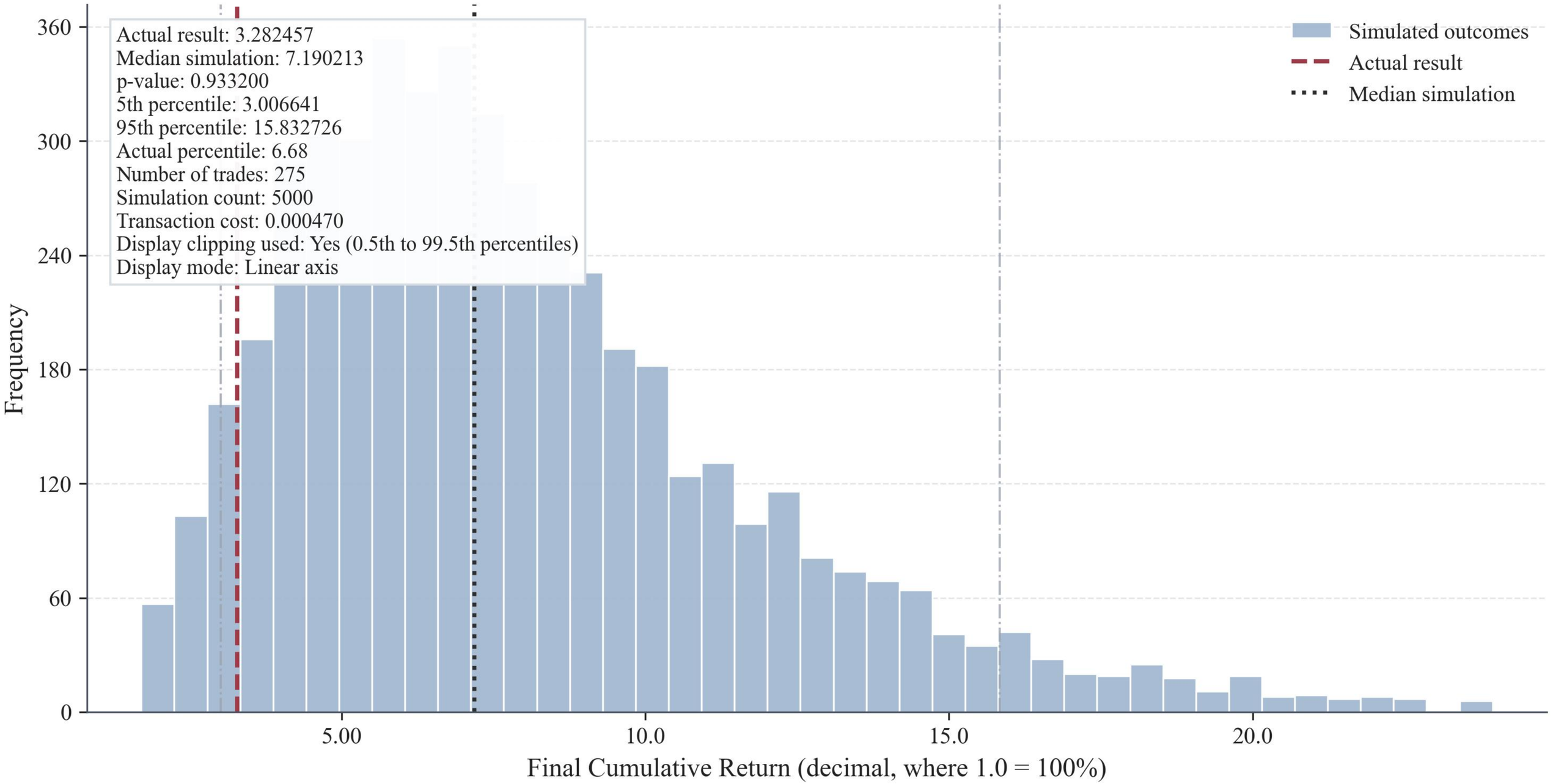
Gray cells are suppressed because the regime cell has fewer than 20 trades.



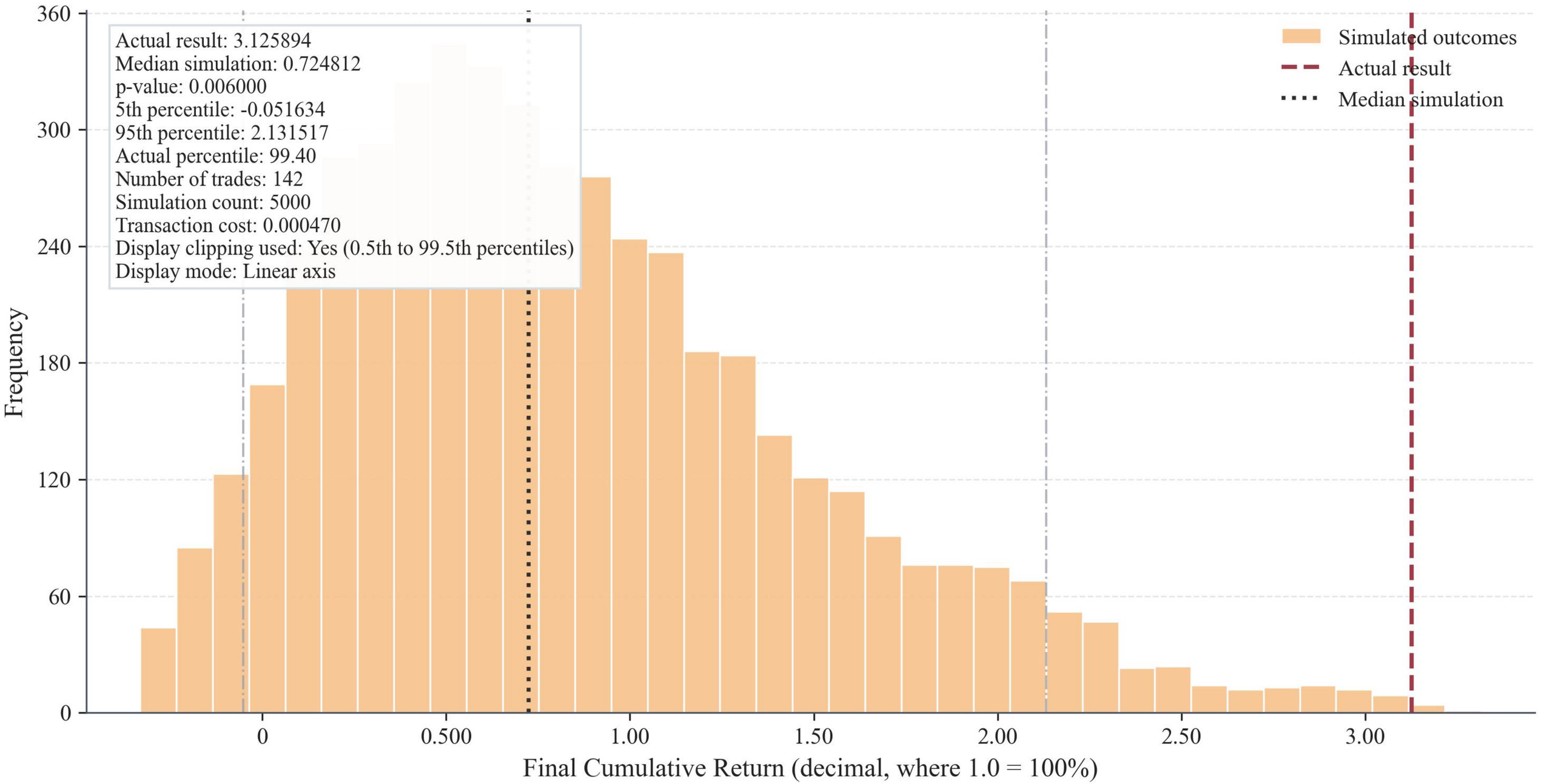
SPY: Daily Equity Curves by Strategy and Benchmark



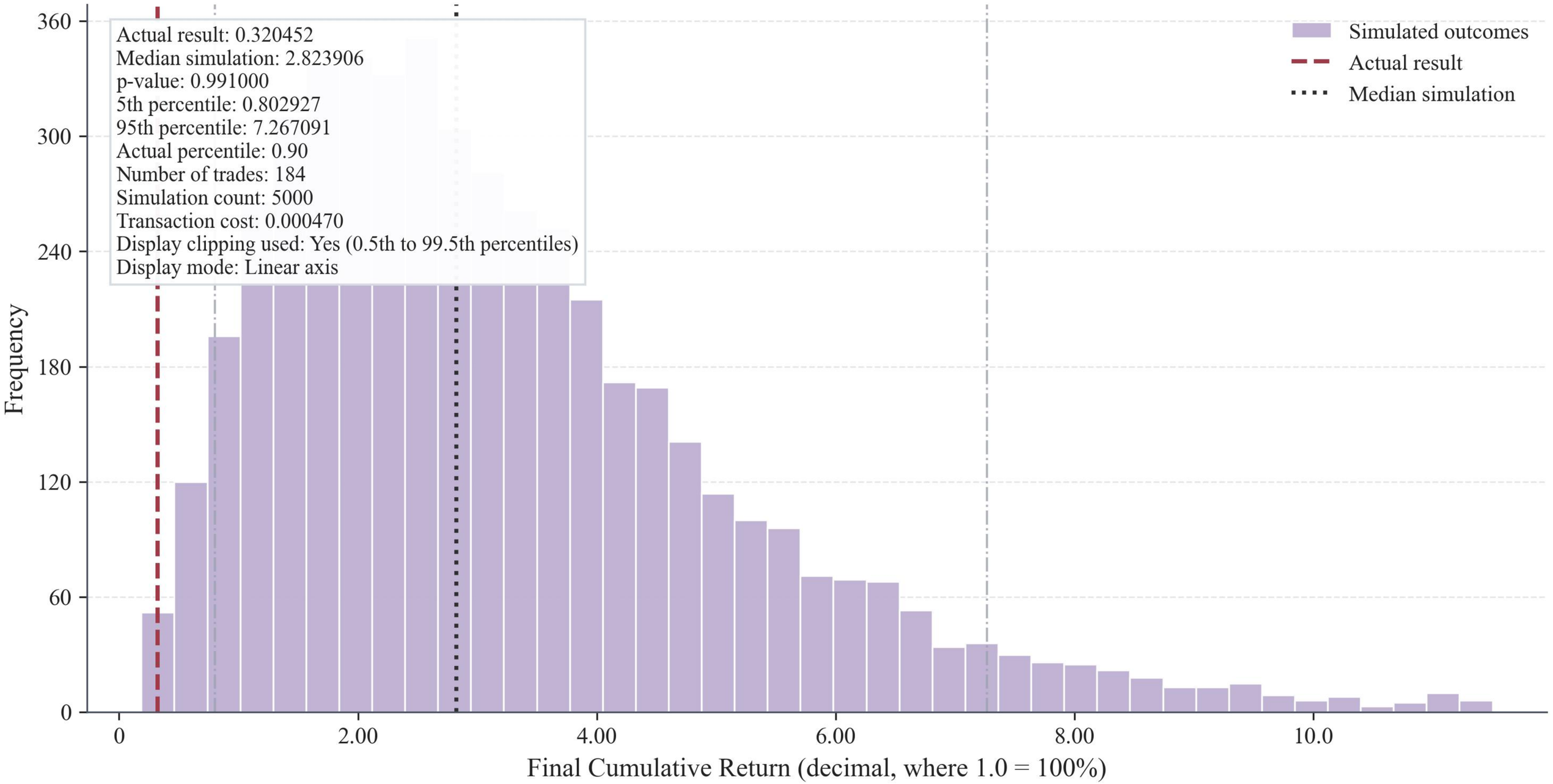
SPY: Monte Carlo Distribution of Final Returns, Trend Strategy

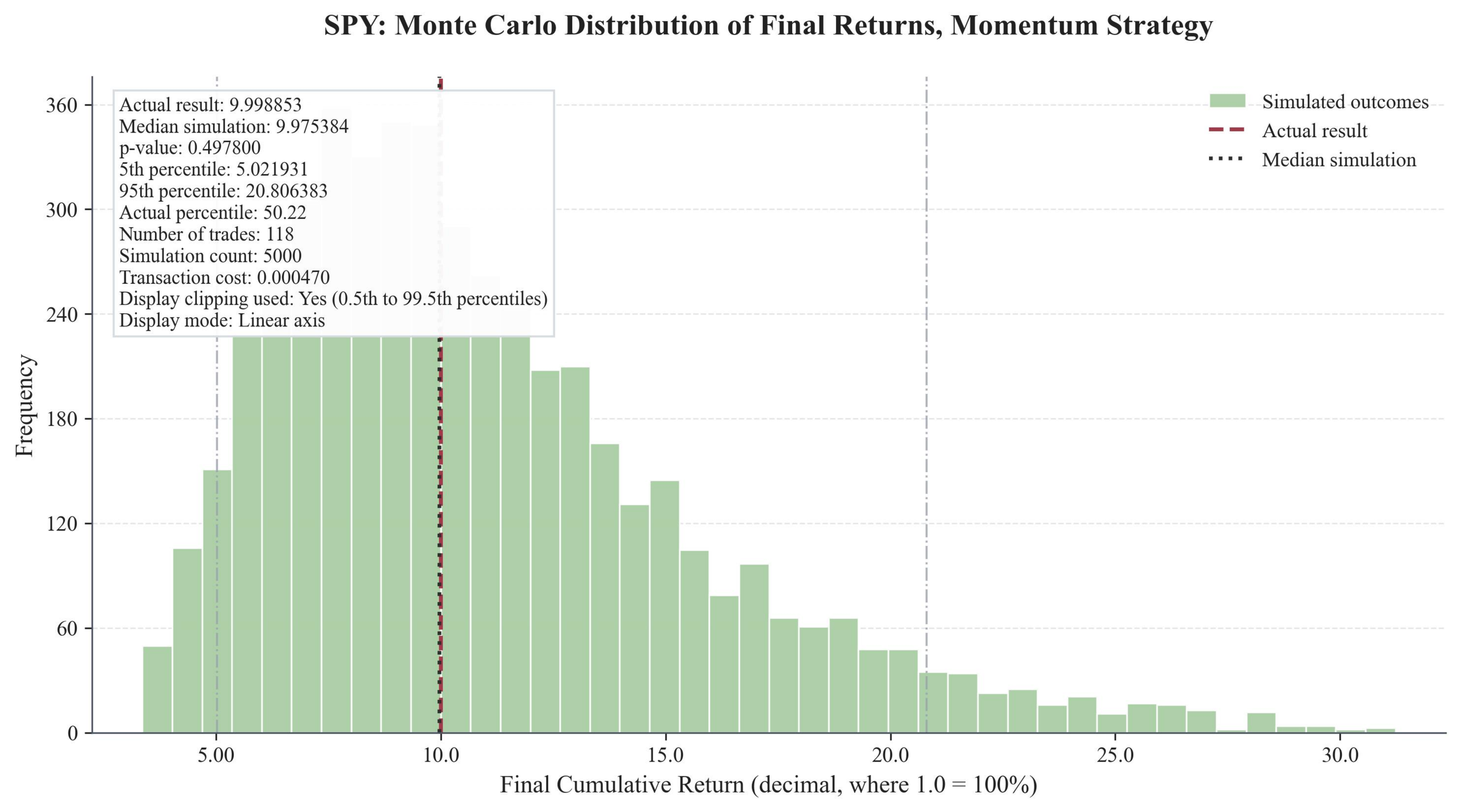


SPY: Monte Carlo Distribution of Final Returns, Mean Reversion Strategy

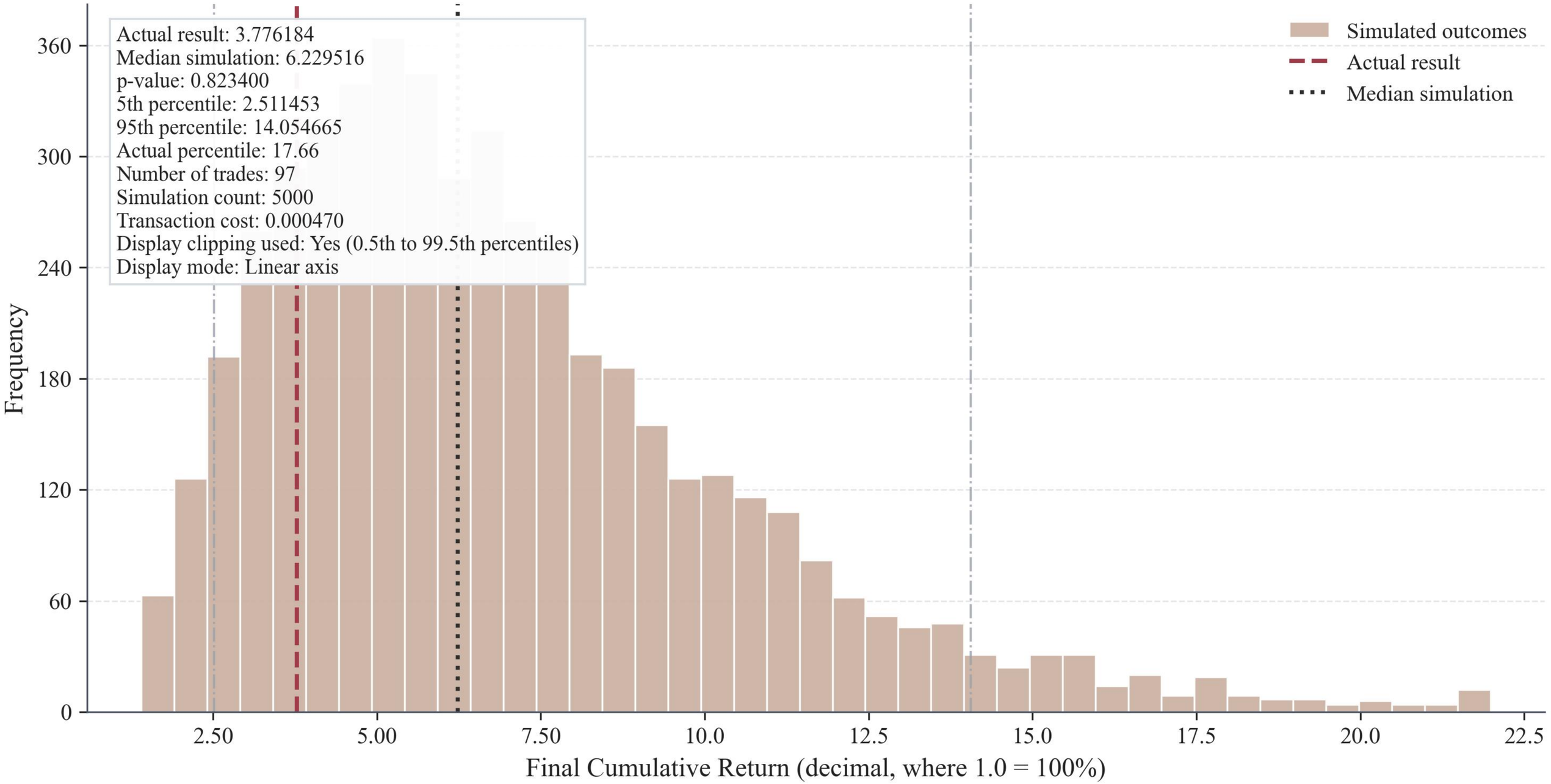


SPY: Monte Carlo Distribution of Final Returns, Random Strategy

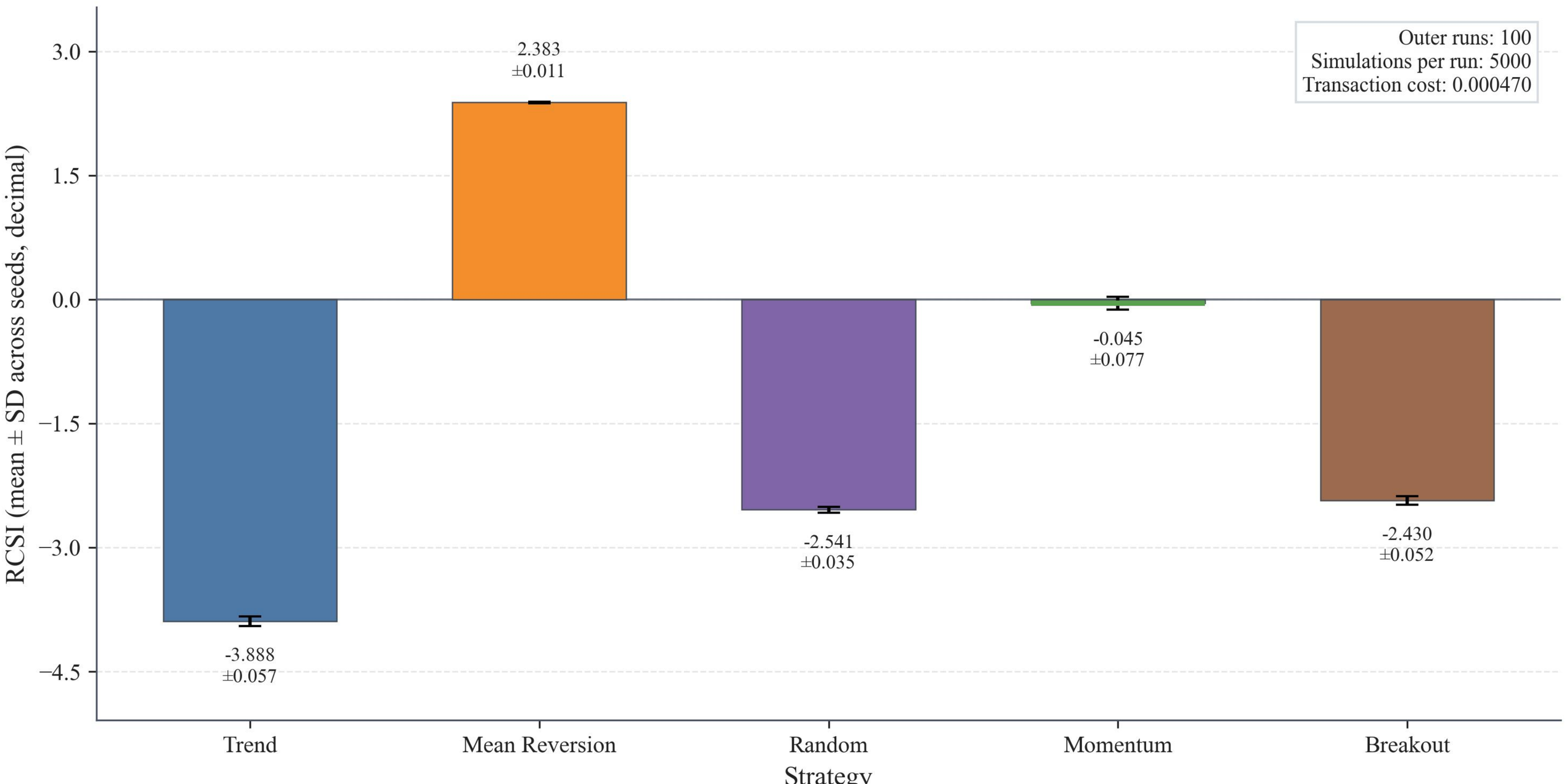



SPY: Monte Carlo Distribution of Final Returns, Momentum Strategy
Actual result: 9.998853
Median simulation: 9.975384
p-value: 0.497800
5th percentile: 5.021931
95th percentile: 20.806383
Actual percentile: 50.22
Number of trades: 118
Simulation count: 5000
Transaction cost: 0.000470
Display clipping used: Yes (0.5th to 99.5th percentiles)
Display mode: Linear axis
Simulated outcomes
Actual result
Median simulation
Frequency
0
60
120
180
240
300
360
5.00
10.0
15.0
20.0
25.0
30.0
Final Cumulative Return (decimal, where 1.0 = 100%)

SPY: Monte Carlo Distribution of Final Returns, Breakout Strategy

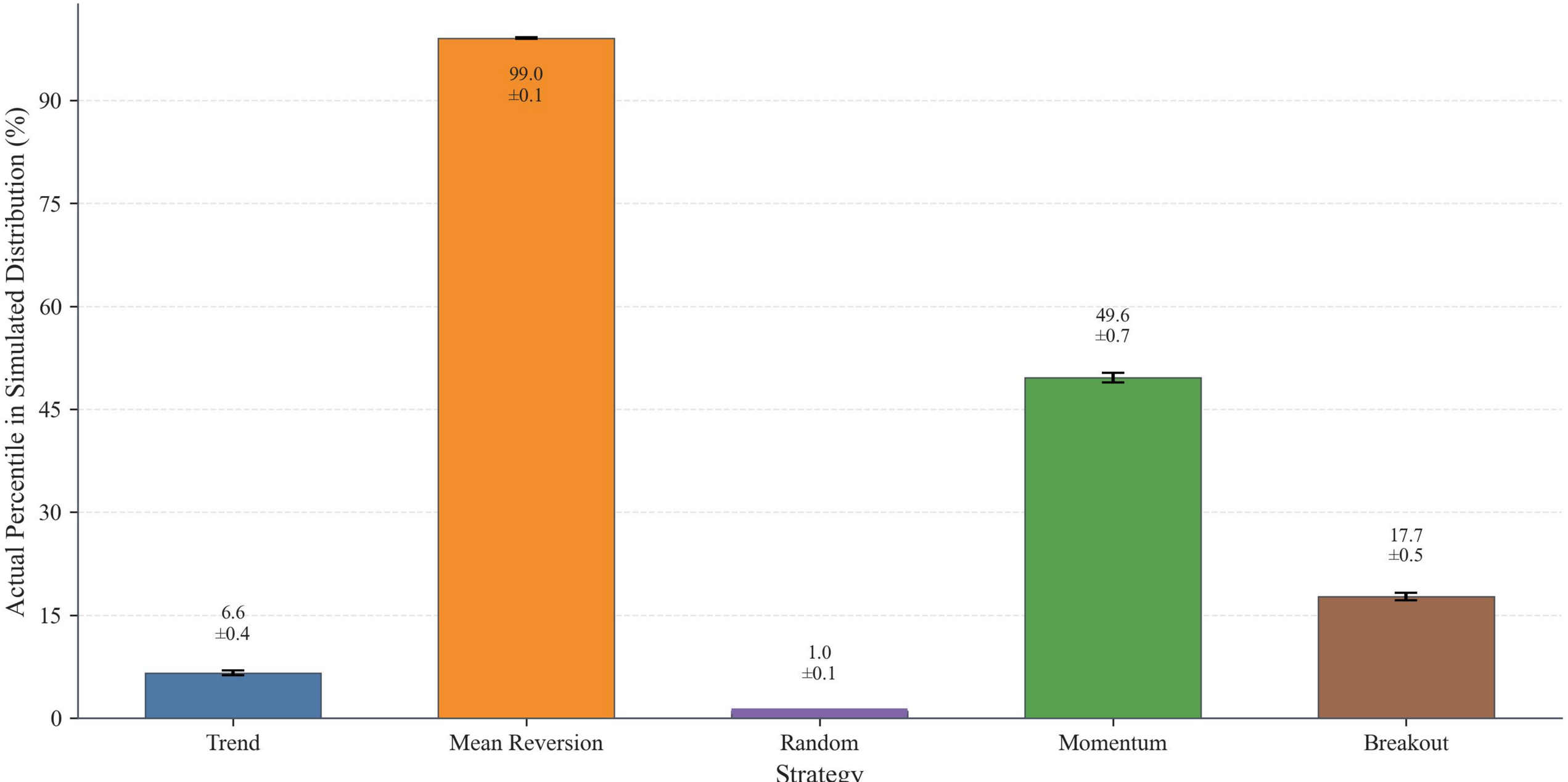


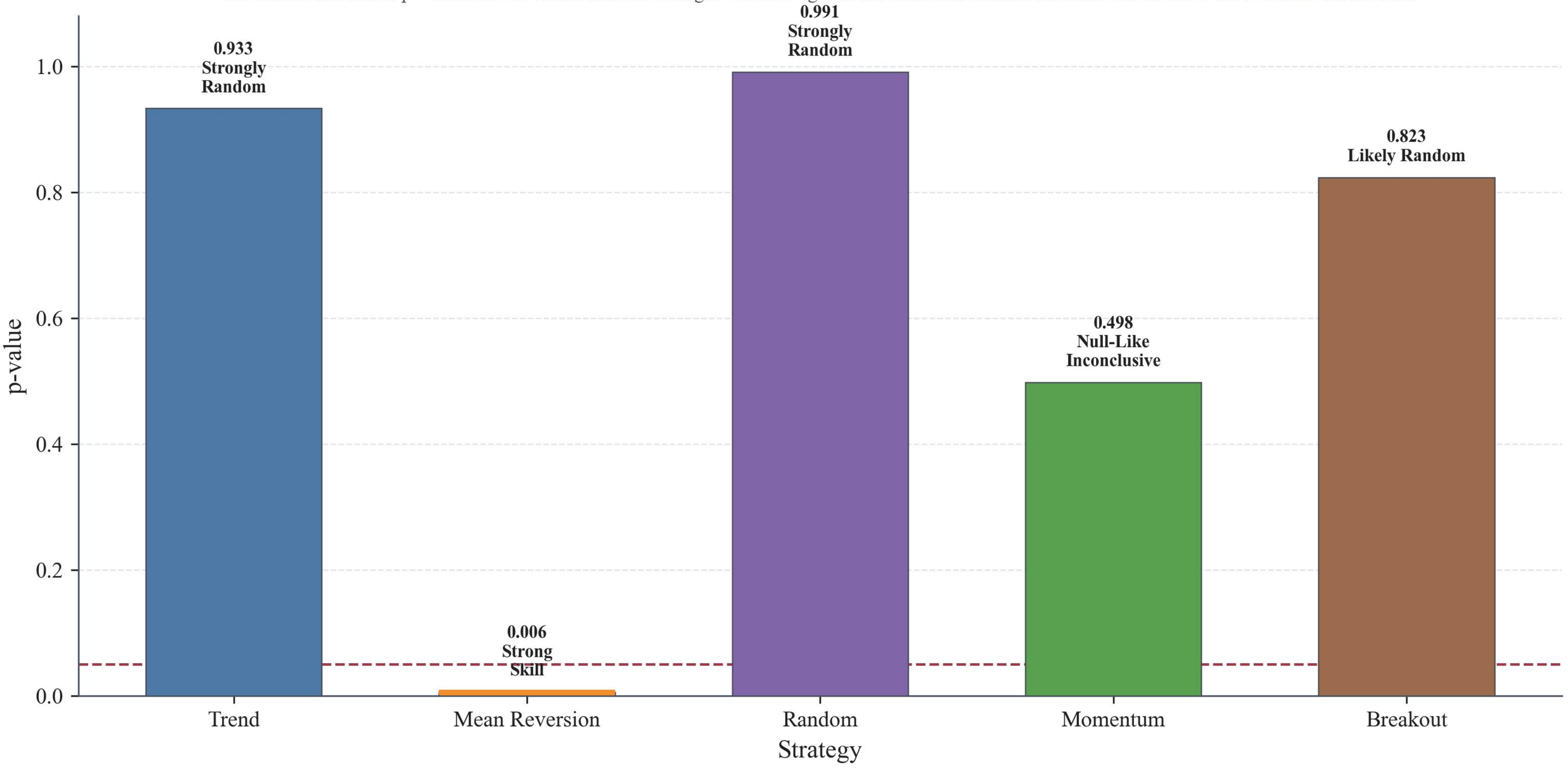
SPY: Monte Carlo Robustness of RCSI Across Seeds



SPY: Monte Carlo Robustness of Actual Percentile Across Seeds

Higher values place the observed outcome deeper into the right tail of the simulated distribution.



SPY: Monte Carlo p-Value by Strategy

The dashed line marks p = 0.05. Lower values indicate stronger evidence against the simulated-random baseline. Labels show the overall evidence class.